

browser fingerprinting

Younes Menfalouti

Code

```
fingerprintingLab > app.py > ...
1  from flask import Flask, render_template, request
2
3  app = Flask(__name__)
4
5
6  @app.route("/")
7  def home():
8      print("\n===== NEW VISIT =====")
9      print("IP address :", request.remote_addr)
10     print("HTTP method :", request.method)
11     print("User-Agent :", request.headers.get("User-Agent"))
12     print("Accept-Language :", request.headers.get("Accept-Language"))
13     print("Accept :", request.headers.get("Accept"))
14     print("Accept-Encoding :", request.headers.get("Accept-Encoding"))
15     print("Referer :", request.headers.get("Referer"))
16     print("Sec-GPC :", request.headers.get("Sec-GPC"))
17     print("=====\n")
18     return render_template("index.html")
19
20
21 @app.route("/collect", methods=["POST"])
22 def collect():
23     print("Received :", request.get_json())
24     return ""
25
26
27 if __name__ == "__main__":
28     app.run(host="127.0.0.1", port=9000, debug=True)
29
```

Figure 1. The Flask server app.py.

```
fingerprintingLab > templates > index.html > html > body > p#target-sentence
1  <!DOCTYPE html>
2  <html>
3
4  <body>
5      <h2>Active Feature Collection</h2>
6      <pre id="feature-output">Waiting for JavaScript...</pre>
7
8      <p id="fingerprint-output"></p>
9
10     <h2>The Typing Imposter</h2>
11     <p id="target-sentence">Cookies are gone, but Alice leaves traces</p>
12     <input id="typing-input" size="60">
13     <button id="restart-button">Restart</button>
14
15     <p id="typing-time"></p>
16     <p id="typing-speed"></p>
17     <p id="typing-corrections"></p>
18
19     <script src="{{ url_for('static', filename='fingerprint.js') }}"></script>
20     <script src="{{ url_for('static', filename='fingerprint2.js') }}"></script>
21 </body>
22
23 </html>
24
```

Figure 2. The page index.html.

Passive features

Passive features arrive with every request, active features need a script in the page, and behavioural features come from how the user interacts with it.

Without any JavaScript the server learns the IP address, the browser and its version, the operating system, the languages, the accepted formats and encodings, and the privacy preference through Sec-GPC.

```
===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent : Mozilla/5.0 (X11; Linux x86_64; rv:151.0) Gecko/20100101 Firefox/151.0
Accept-Language : en-US,en;q=0.9
Accept : text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, zstd
Referer : None
Sec-GPC : None
=====

127.0.0.1 - - [02/Oct/2026 22:44:49] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:44:49] "GET /favicon.ico HTTP/1.1" 404 -

===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent : Mozilla/5.0 (X11; Linux x86_64; rv:151.0) Gecko/20100101 Firefox/151.0
Accept-Language : en-US,en;q=0.9
Accept : text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, zstd
Referer : None
Sec-GPC : None
=====
```

```
=====
127.0.0.1 - - [02/Oct/2026 22:50:32] "GET / HTTP/1.1" 200 -

===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent : Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36
Accept-Language : en-US,en;q=0.9
Accept : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, zstd
Referer : None
Sec-GPC : 1
=====

127.0.0.1 - - [02/Oct/2026 22:57:18] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:57:18] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 - - [02/Oct/2026 22:57:18] "GET /static/fingerprint.js HTTP/1.1" 304 -
127.0.0.1 - - [02/Oct/2026 22:57:18] "GET /static/fingerprint.js HTTP/1.1" 304 -
Received : {'fp': 'd426d30f50d0980e3e7bcd92afbac9ee0290f62c7bf04831570068f7ca4fcfe'}
127.0.0.1 - - [02/Oct/2026 22:57:18] "POST /collect HTTP/1.1" 200 -
Received : {'language': 'en-US', 'cores': 16, 'screen': '1280x800', 'window': '636x759', 'timezone': 'Africa/Casablanca'}
127.0.0.1 - - [02/Oct/2026 22:57:18] "POST /collect HTTP/1.1" 200 -
```

Figure 3. Passive features of Firefox and Brave.

The language, the encodings, the IP and the operating system stay the same in both browsers. The User-Agent, the Accept header and Sec-GPC change. Brave also sends the client hints Sec-Ch-Ua and Sec-Ch-Ua-Platform. The Firefox private window sends the same headers except Sec-GPC, which becomes 1.

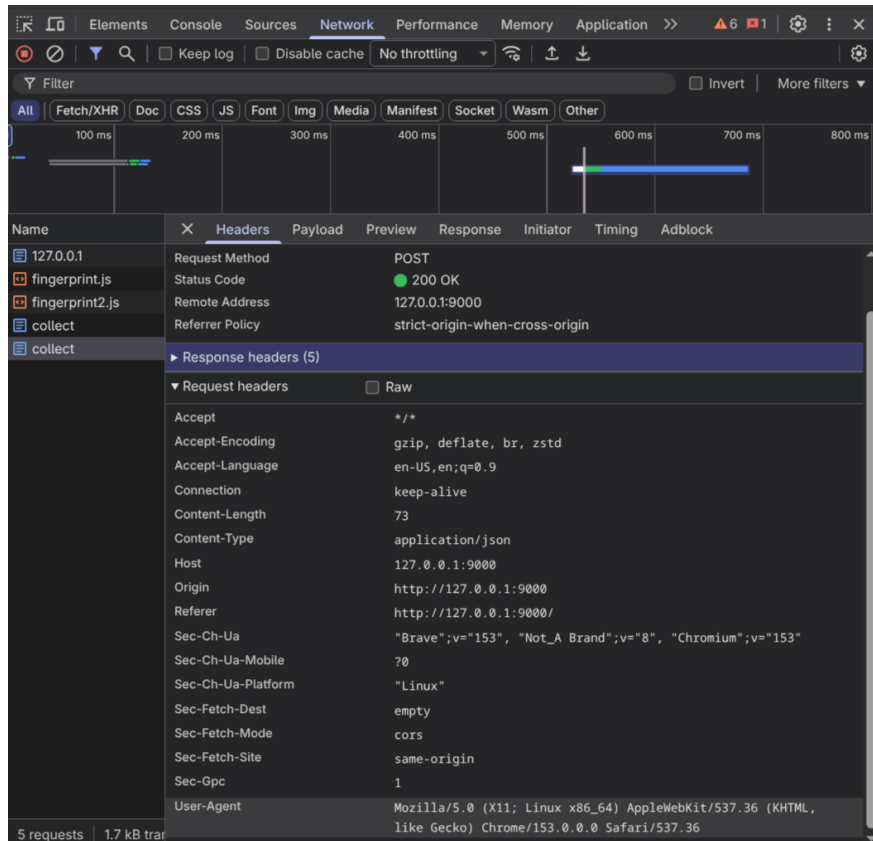


Figure 4. Client hints sent by Brave.

```

===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent : Mozilla/5.0 (X11; Linux x86_64; rv:151.0) Gecko/2
0100101 Firefox/151.0
Accept-Language : en-US,en;q=0.9
Accept : text/html,application/xhtml+xml,application/xml;q=0.9
,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, zstd
Referer : None
Sec-GPC : 1
=====

127.0.0.1 - - [02/Oct/2026 22:57:55] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:57:55] "GET /favicon.ico HTTP/1.
1" 404 -

```

Figure 5. Firefox private window.

Active features

The language reveals the user settings, the number of cores the processor, the screen size the monitor, the window size the browser layout and the time zone the region. Each value alone is common, but together they narrow down the device.

```

fingerprintingLab > static > JS fingerprint.js > ...
1  const features = {
2    language: navigator.language,
3    cores: navigator.hardwareConcurrency,
4    screen: screen.width + "x" + screen.height,
5    window: window.innerWidth + "x" + window.innerHeight,
6    timezone: Intl.DateTimeFormat().resolvedOptions().timeZone
7  };
8
9  document.getElementById("feature-output").textContent = JSON.stringify(features, null, 2);
10
11  fetch("/collect", { method: "POST", headers: { "Content-Type": "application/json" }, body: JSON.stringify(features) });
12
13  // Fingerprint: SHA-256 of the features joined in a fixed order
14  const combined = Object.values(features).join(" | ");
15
16  crypto.subtle.digest("SHA-256", new TextEncoder().encode(combined)).then(function (buffer) {
17    const hash = Array.from(new Uint8Array(buffer)).map(b => b.toString(16).padStart(2, "0")).join("");
18    document.getElementById("fingerprint-output").textContent = "Fingerprint: " + hash;
19    fetch("/collect", { method: "POST", headers: { "Content-Type": "application/json" }, body: JSON.stringify({ fp: hash }) });
20  });
21

```

Figure 6. The script `fingerprint.js`.

Active Feature Collection

```

{
  "language": "en-US",
  "cores": 16,
  "screen": "1440x900",
  "window": "1428x759",
  "timezone": "Africa/Casablanca"
}

```

Fingerprint: d07c0d14ec9806d787510a376b0e02ac3dcd8e99c4243f3d179c4666d41c926d

The Typing Imposter

Cookies are gone, but Alice leaves traces

```

127.0.0.1 - - [02/Oct/2026 22:48:47] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:48:47] "GET /static/fingerprint2.js HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:48:47] "GET /static/fingerprint.js HTTP/1.1" 200 -
Received : {'language': 'en-US', 'cores': 16, 'screen': '1440x900', 'window': '1428x759', 'timezone': 'Africa/Casablanca'}
127.0.0.1 - - [02/Oct/2026 22:48:48] "POST /collect HTTP/1.1" 200 -
Received : {'fp': 'd07c0d14ec9806d787510a376b0e02ac3dcd8e99c4243f3d179c4666d41c926d'}
127.0.0.1 - - [02/Oct/2026 22:48:48] "POST /collect HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:48:48] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [02/Oct/2026 22:49:05] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -

```

Figure 7. Features shown in Brave and received by the server.

With the ad blocker on, Firefox blocks `fingerprint.js` and the server gets no active feature. Without it, both scripts load and both requests reach `/collect`.

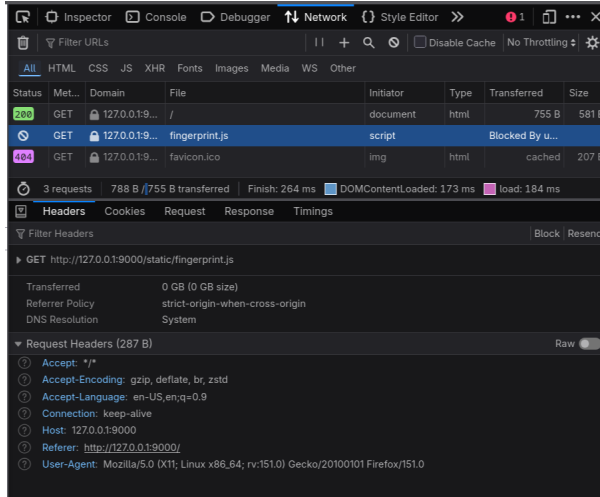
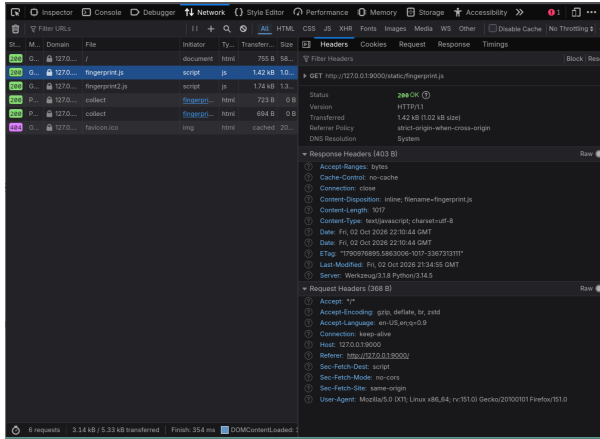


Figure 8. Firefox with the ad blocker.



Active Feature Collection

Waiting for JavaScript...

The Typing Imposter

Cookies are gone, but Alice leaves traces

Figure 9. Firefox without the ad blocker.

Cookie blocking and private browsing change nothing for these features, since none of them is stored in the browser. Blocking the script is the most effective protection here. Its limits are that the passive headers still reach the server and that a renamed script would not be blocked.

Typing behaviour

$$speed = \frac{characters/5}{seconds/60}$$

The script starts the timer on the first key, adds one correction on each Backspace, stops when the input equals the sentence, then computes the speed, displays it and sends it.

```

fingerprintingLab > static > JS fingerprint2.js > ...
1  const sentence = document.getElementById("target-sentence").textContent;
2  const input = document.getElementById("typing-input");
3
4  let startTime = null;
5  let corrections = 0;
6
7  input.addEventListener("keydown", function (event) {
8      if (startTime === null) {
9          startTime = performance.now();
10     }
11     if (event.key === "Backspace") {
12         corrections = corrections + 1;
13     }
14 });
15
16 input.addEventListener("input", function () {
17     if (input.value !== sentence) {
18         return;
19     }
20
21     // Speed in words per minute, one word = 5 characters
22     const seconds = (performance.now() - startTime) / 1000;
23     const speed = (sentence.length / 5) / (seconds / 60);
24
25     document.getElementById("typing-time").textContent = "Time: " + seconds.toFixed(2) + " s";
26     document.getElementById("typing-speed").textContent = "Speed: " + speed.toFixed(2) + " wpm";
27     document.getElementById("typing-corrections").textContent = "Corrections: " + corrections;
28
29     const typing = { time: seconds.toFixed(2), speed: speed.toFixed(2), corrections: corrections };
30     fetch("/collect", { method: "POST", headers: { "Content-Type": "application/json" }, body: JSON.stringify(typing) });
31 });
32
33 document.getElementById("restart-button").addEventListener("click", function () {
34     startTime = null;
35     corrections = 0;
36     input.value = "";
37 });
38

```

Figure 10. The script `fingerprint2.js`.

The first run takes 8.35 s at 58.93 wpm with no correction. The second takes 122.62 s at 4.01 wpm with 52 corrections. The same user gives very different values, so typing alone is a weak identifier.

Active Feature Collection

```

{
  "language": "en-US",
  "cores": 16,
  "screen": "1280x800",
  "window": "636x759",
  "timezone": "Africa/Casablanca"
}

```

Fingerprint: d426d30f50d0980e3e7bcd92afbac9ee0290f62c7bf04831570068f7ca4fcfe

The Typing Imposter

Cookies are gone, but Alice leaves traces

Cookies are gone, but Alice leaves traces

Restart

Time: 8.35 s

Speed: 58.93 wpm

Corrections: 0

The screenshot shows the Chrome DevTools Network tab. A POST request to `/collect` is selected. The 'Request payload' tab is active, showing the following JSON data:

```

{
  "time": "8.35",
  "speed": "58.93",
  "corrections": 0
}

```

Figure 11. First run and its payload.

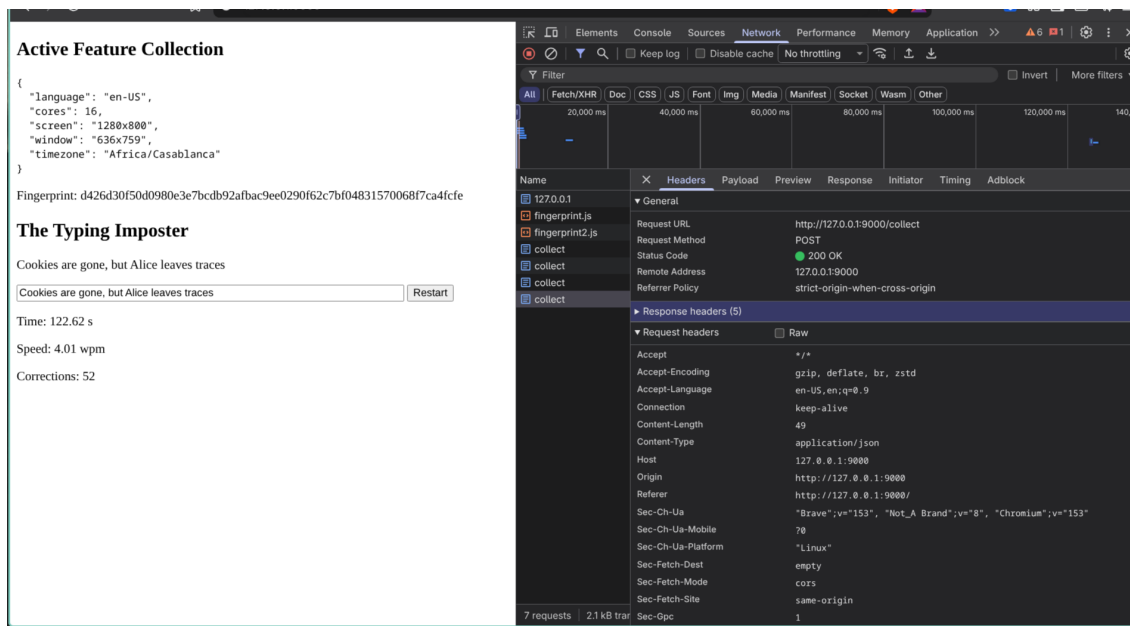


Figure 12. Second run.

Fingerprint

The same features give the same SHA-256 hash on every reload. A change of screen or window size changes the hash from d07c0d14... to d426d30f.... The fingerprint recognises the browser without cookies, but one changed feature is enough to break it.

```
127.0.0.1 - - [02/Oct/2026 22:49:09] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:49:09] "GET /static/fingerprint2.js HTTP/1.1" 304 -
127.0.0.1 - - [02/Oct/2026 22:49:09] "GET /static/fingerprint.js HTTP/1.1" 304 -
127.0.0.1 - - [02/Oct/2026 22:49:09] "GET /.well-known/appspec ific/com.chrome.devtools.json HTTP/1.1" 404 -
Received : {'language': 'en-US', 'cores': 16, 'screen': '1280x800', 'window': '636x759', 'timezone': 'Africa/Casablanca'}
127.0.0.1 - - [02/Oct/2026 22:49:09] "POST /collect HTTP/1.1" 200 -
Received : {'fp': 'd426d30f50d0980e3e7bcd92afbac9ee0290f62c7bf04831570068f7ca4fcfe'}
127.0.0.1 - - [02/Oct/2026 22:49:09] "POST /collect HTTP/1.1" 200 -

===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent : Mozilla/5.0 (X11; Linux x86_64; rv:151.0) Gecko/20100101 Firefox/151.0
Accept-Language : en-US,en;q=0.9
Accept : text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, zstd
Referer : None
Sec-GPC : None

127.0.0.1 - - [02/Oct/2026 22:50:32] "GET / HTTP/1.1" 200 -
```

```
127.0.0.1 - - [02/Oct/2026 22:57:18] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Oct/2026 22:57:18] "GET /.well-known/appspec ific/com.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 - - [02/Oct/2026 22:57:18] "GET /static/fingerprint.js HTTP/1.1" 304 -
127.0.0.1 - - [02/Oct/2026 22:57:18] "GET /static/fingerprint2.js HTTP/1.1" 304 -
Received : {'fp': 'd426d30f50d0980e3e7bcd92afbac9ee0290f62c7bf04831570068f7ca4fcfe'}
127.0.0.1 - - [02/Oct/2026 22:57:18] "POST /collect HTTP/1.1" 200 -
Received : {'language': 'en-US', 'cores': 16, 'screen': '1280x800', 'window': '636x759', 'timezone': 'Africa/Casablanca'}
127.0.0.1 - - [02/Oct/2026 22:57:18] "POST /collect HTTP/1.1" 200 -
```

Figure 13. A new hash after the size change, then the same hash on reload.